

Amrith Akshintala

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OBJECTIVE

Rising Sophomore Engineering Honors student at Texas A&M with hands-on experience in embedded systems, robotics, and software development through project teams and research. Eager to develop practical autonomous systems and real-time software development with mentorship from experienced engineers.

EDUCATION

Texas A&M University — College Station, TX

May 2029

- B.S. General Engineering (Intended Computer Engineering after ETAM) | GPA: 3.43 | Engineering Honors

TECHNICAL PROJECTS

2-Axis Gimbal Project

May 2026 – Present

- Designing a 2-axis camera gimbal with a team of 4 using Arduino GY-87 IMU and SG90 servos; implemented PID stabilization using C++
- Completed full CAD modeling and assembly in SolidWorks; currently in final wiring and hardware assembly

StudyBot — Personal Project

June 2026 – Present

- Built and deployed a full-stack AI-powered study assistant using React, Node.js/Express, and the Anthropic Claude API
- Implemented workspace-based organization, PDF text extraction, prompt caching (reducing per-message API cost by ~90%), and an interactive multiple-choice quiz system with real-time grading and AI-generated explanations

Disease Classification from Medical Images — Aggie Data Science Club

Jan 2026 – Present

- Built a ResNet-50 deep learning model to classify chest X-rays into 3 categories (Normal, Bacterial Pneumonia, Viral Pneumonia), achieving 86.4% validation accuracy
- Performed data preprocessing, augmentation, and evaluation using PyTorch, NumPy, and Pandas

LegalDefender — Hackathon

2026

- Built an AI-powered tenant protection app using Gemini 1.5 Pro to scan lease PDFs against Texas Property Code and flag illegal clauses in real-time; presented at TAMUHack hackathon
- Implemented decentralized landlord reputation tracking via verified reviews minted to the Solana blockchain; built with Next.js, React, and PostgreSQL

Cosmos — Hackathon

2026

- Built an interactive force-directed knowledge graph visualizing NASA exoplanet and mission data, with an AI guide powered by the Claude API; Developed in React and D3.js; presented at Build4Good hackathon

RELEVANT EXPERIENCE

Member—Texas A&M TURTLE Robotics

Sep 2025 – Dec 2025

- Developed foundational skills in Python, C++, electronics, CAD modeling, circuits, and manufacturing
- Collaborated with a team to design, build, and test a robot a block-retrieval challenge against opposing robots; designed all components in SolidWorks

Research Intern — Texas A&M Ultrasound Research Laboratory

Aug 2025 – Present

- Developed non-invasive ultrasound techniques to measure tumor pressure

Research Intern — Dartmouth-Hitchcock Medical Center

Jun 2024 – Aug 2024

- Conducted biomedical research using computational models for disease diagnosis and pathological data analysis

VOLUNTEER EXPERIENCE

Co-founder — Scholar Sphere

Jul 2023 – Jul 2025

- Established a nonprofit providing online educational resources to high school students
- Developed an online SAT practice platform with full-length test simulation using HTML, CSS, and JavaScript
- Established partnerships with 22 nonprofits; earned the PVSA Gold Award for volunteer service

Lead Tutor — Voices Academy

Nov 2021 – Dec 2023

- Led virtual tutoring sessions for K–6 students from low-income backgrounds

SKILLS

Technical: Python, Java, JavaScript, HTML, CSS, React, Swift, SystemVerilog, TensorFlow, PyTorch, SolidWorks, Research, Lab techniques

Professional: Leadership, Teamwork, Time Management, Problem-Solving, Communication

AWARDS & HONORS

Eagle Scout | AP Scholar with Distinction | National Merit Commended Scholar | PVSA Gold Award